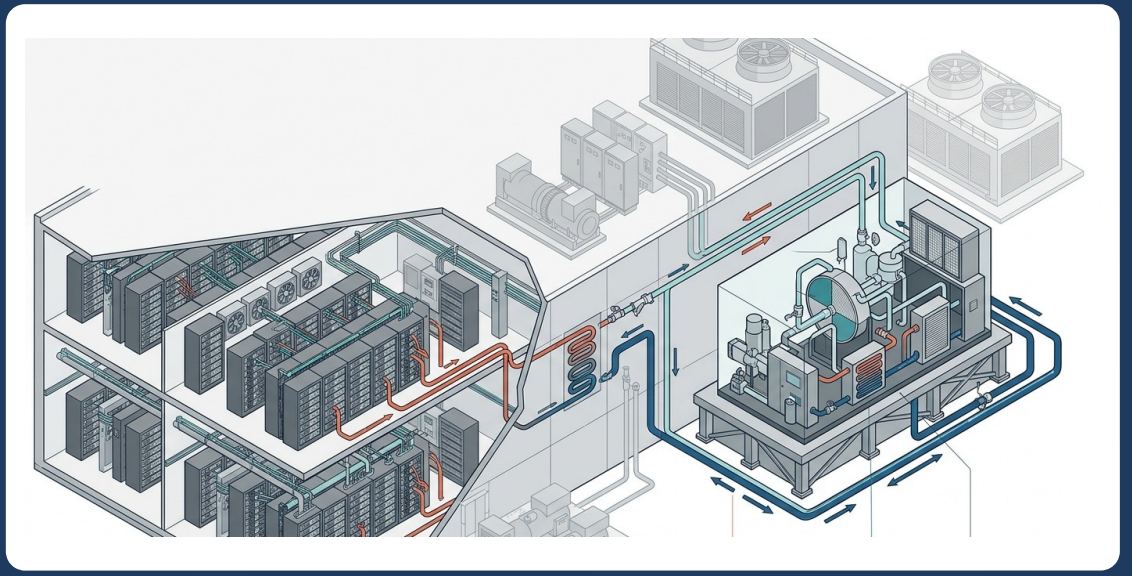
AirTransformer

Heat-driven cooling for climate-resilient data centres

EIC Accelerator 2026 · Challenge C05

Deep Tech for Climate Adaptation

Savery Tech, France · Frédéric Cui, founder and CEO



COP 20

measured with 60 °C waste heat, 2025 EN test

TRL6 → TRL8

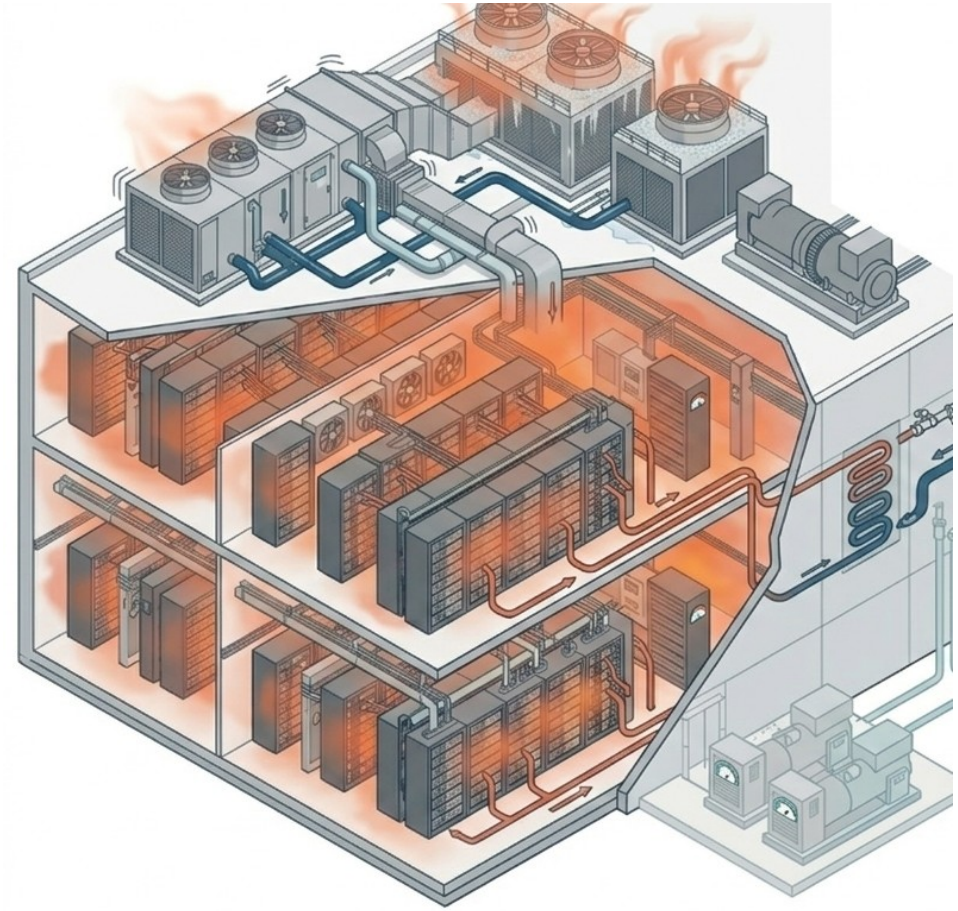
qualified pre-series product by March 2029

€4.49m EIC

€2.49m grant + €2m equity in an €8m Seed

Heatwaves push cooling into grid and water limits

Electricity and water demand peak exactly when both are scarcest, while low-grade heat goes unused.



EER ≈ 3

In a heatwave, compression cooling draws 333 kW of electricity per MW of cooling, at the hours of peak grid stress.

Power-limited

Q1 2026 inventory grew 33% in the four largest North American markets and 18.9% in Europe; power availability now limits where sites go (CBRE, June 2026).

Water

Evaporative cooling needs make-up water that competes with cities when drought restrictions apply.

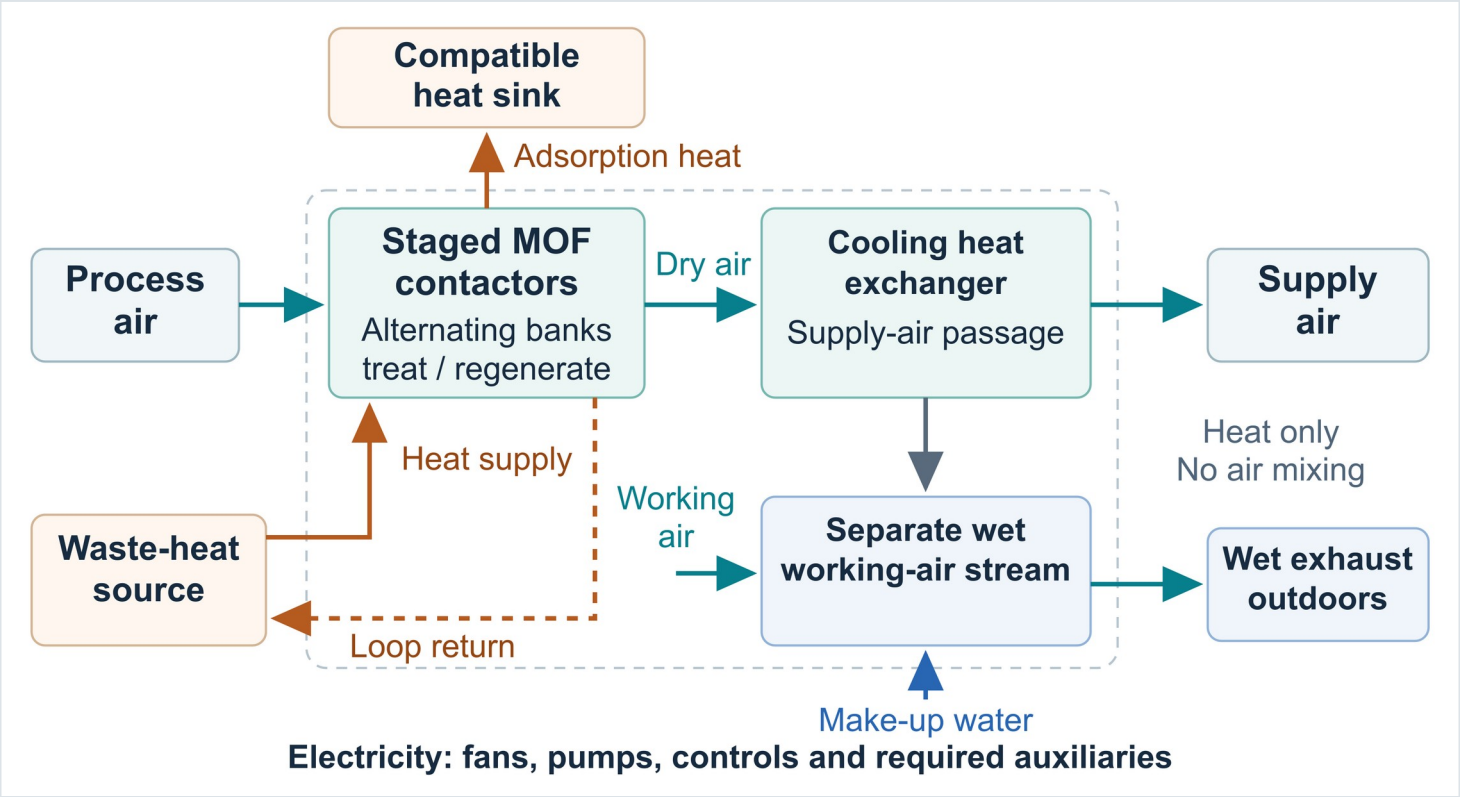
PUE ≤ 1.2

Germany's EnEfG requires data centres entering service from July 2026 to reach it within two years.

Meanwhile, liquid-cooled racks reject 40–70 °C heat that nobody uses.

Rack heat in, cooling out: no vapour compression

Waste heat dries the air in staged MOF contactors; a cooling heat exchanger delivers the cooling.



Regenerates at 50–70 °C with water recovery and 40–50 °C without. Installs beside existing cooling; the site keeps its isolation, fallback and supervision.

COP 20

machine electrical COP with 60 °C waste heat (2025 EN test); 8–20 installed

−60–70%

electricity at equal cooling: action qualification target

−40–60%

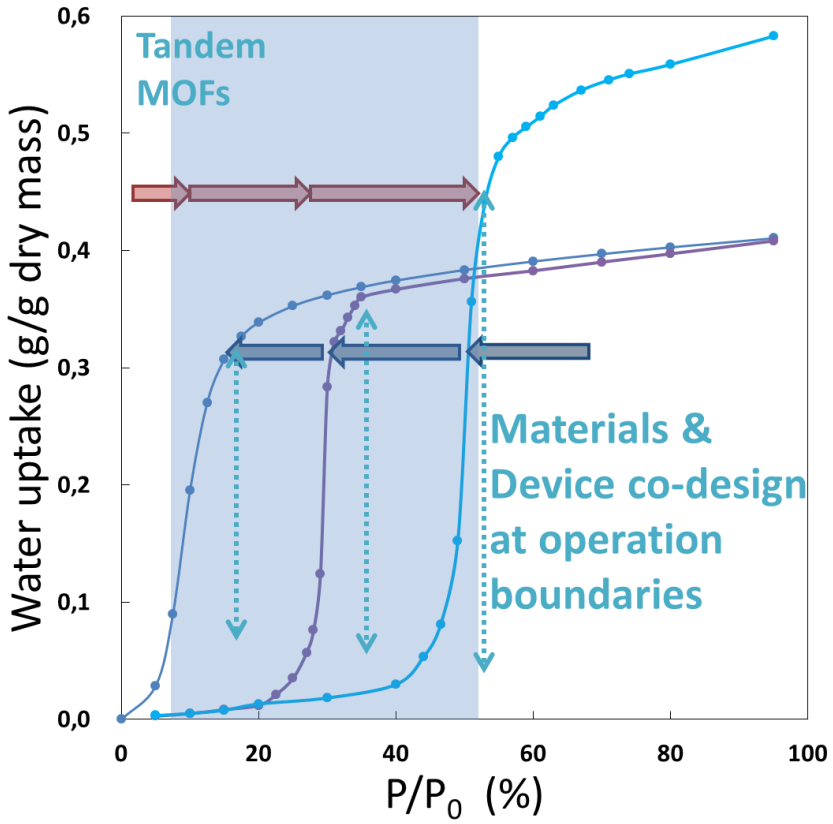
external make-up water at equal cooling

20 → 600 kW

20 kW thermal module, 600 kW container configuration

Material, cycle and machine engineered as one

Molecular sorption becomes sustained cooling only when contactor, heat exchange and timing are co-designed.



Tandem MOF isotherms: uptake steps placed at the machine's operating boundaries.

1

Material

Aluminium-based CAU-family MOFs (Kiel University) take up and release water in steep, low-temperature steps; each stage uses a different window as the air dries.

2

Contactor

Coated fins and fluid-carrying tubes remove adsorption heat as it forms. This staged arrangement is protected by the 2025 family FR2508386.

3

Cycle

Finite-time design sets switching, regeneration and heat recovery for useful cooling per cycle, building on Henninger et al., co-authored by F. Cui.

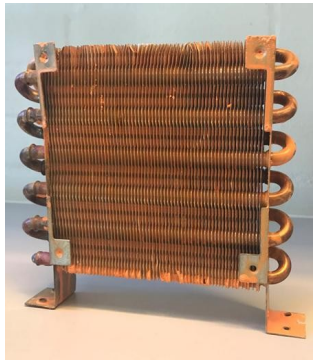
4

Control

Model-predictive supervision inside independent safety limits and deterministic fallback; operators keep override.

Ten years of prototypes, one complete TRL6 machine

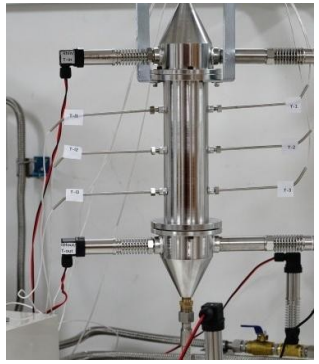
2025 EN test: sensible-cooling electrical COP 20 with 60 °C waste heat.



PoC 1 · 2017



1 kW prototype · 2019



PoC 2 · 2021



PoC 3 · 2022



15 kW prototype · 2023



25 kW prototype · 2024

COP 20

complete machine, EN test, 60 °C heat

TRL6 machine · 2025

Machine COP map at 60 °C source: ambient temperature × relative humidity

	20 °C	25 °C	30 °C	35 °C	40 °C
RH 30%	18	16	16	16	14
RH 40%	18	18	17	16	16
RH 50%	18	17	16	16	15
RH 60%	19	19	18	17	17
RH 70%	18	18	16	16	16

14–19 COP across 20–40 °C ambient and 30–70% RH: performance holds in the heat

Fans, pumps, controls are the only electrical inputs; no compressor in the cooling cycle

8–20 installed Site comparisons include the circulating pump and every affected auxiliary, which the EN test boundary excluded

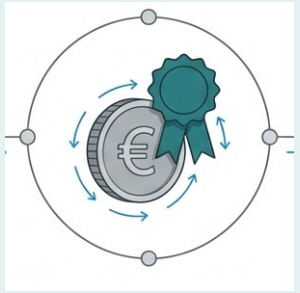
What changed since the 2025 evaluation

Each change answers a point raised by the 2025 evaluators.



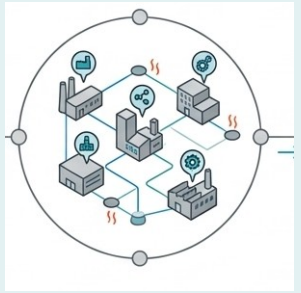
Intellectual property

All three patent families filed by Savery Tech: FR2508386 with its PCT extension, EP26301484.7 (October 2026) and EP3825627A1. Product-specific FTO maps BASF, NuMat, Framergy, AirJoule and Kiel/Fraunhofer process rights.



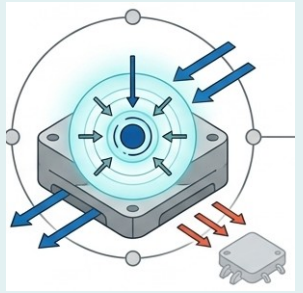
Funding and research

€400k raised through BSA-AIR. CETP 2026–2029 research programme awarded, with HPE Labs contributing data-centre integration expertise.



Team and governance

Two engineers and an interim CFO in post. Board with two Corps des Mines engineers and a data-centre executive; scientific board with Kiel University and CNRS IRCELYON. Target: ≥40% women among all new hires by March 2029.



Product focus

One platform for urban liquid-cooled data centres in Europe and North America. Complete TRL6 machine, COP 20 measured in 2025. Four reference units planned with two prospective customers in 2027.

Savery owns its IP; FTO follows the CAU route

Patents protect the staged exchanger and system architecture; know-how protects the process.

Family (applicant Savery Tech)	Filing	What it protects
Heat-and-mass exchanger, 2025	FR2508386, 21 July 2025; PCT filed 20 July 2026	Staged MOFs with decreasing water-adsorption thresholds, coated fins and integrated heat-transfer tubes: the core of the product
Sorption heat pump, 2026	EP26301484.7, 9 October 2026	Routable vapour circulation with compression assist: the heating extension (COP 6–10)
System family, 2019	EP3825627A1, pending	Earlier system architecture

Know-how kept in-house

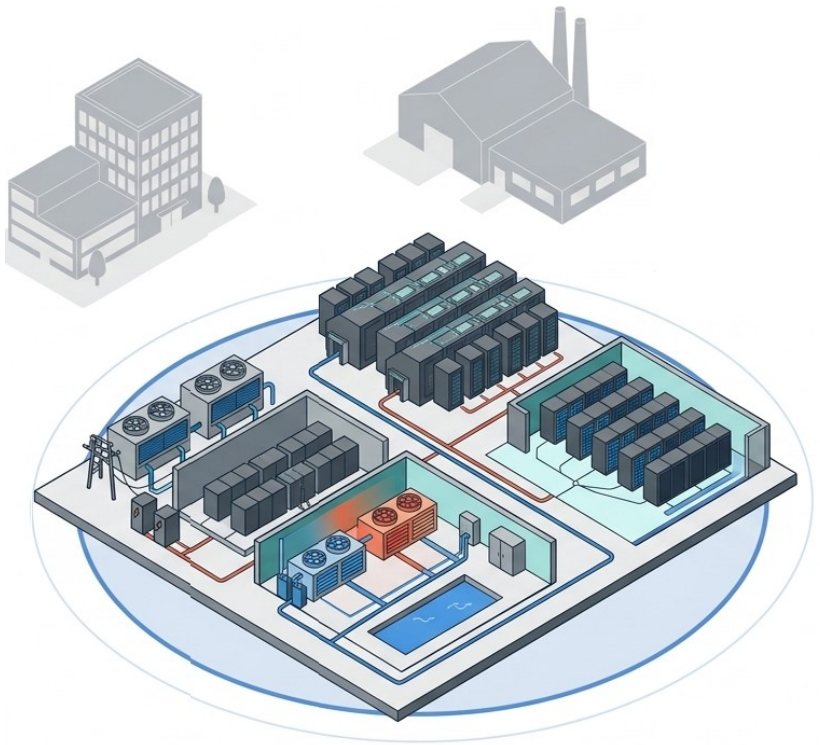
Coating windows, tolerances, calibration, release tests and control parameters stay trade secrets. Supplier agreements define permitted use, improvements and continuing access for production and replacement supply.

Product-specific freedom to operate

1. Screening leads mapped: BASF, NuMat, Framergy, Montana/AirJoule; Kiel/Fraunhofer MOF-process patents EP3350151B1 and US10696704B2.
2. Specialist review compares the selected CAU synthesis or supply route with each claim, by jurisdiction.
3. Disposition (supply or licence, design change, or territory) recorded at the M12 gate, before manufacturing transfer and each release.
4. PCT extension and follow-on filings track the European and North American manufacturing and sales plan.

A €15bn market, entered through liquid-cooled sites

Beachhead: urban liquid-cooled data centres with 50–70 °C rack heat, in Europe and North America.



€15bn

data-centre cooling opportunity, company planning estimate

USD 20.5bn

worldwide data-centre cooling, 2025 (Global Market Insights)

€75m → €181m

Savery revenue 2031 → 2032: 0.5% → 1.2% of the €15bn reference

Entry sequence

2027

Four reference units with two prospective customers

2028–2032

Liquid-cooled data centres in Europe and North America: 1,498 units in 2032

Next

Industrial 40–50 °C waste heat and urban critical facilities, via the 600 kW container

Cooling from waste heat, without compressor or vacuum

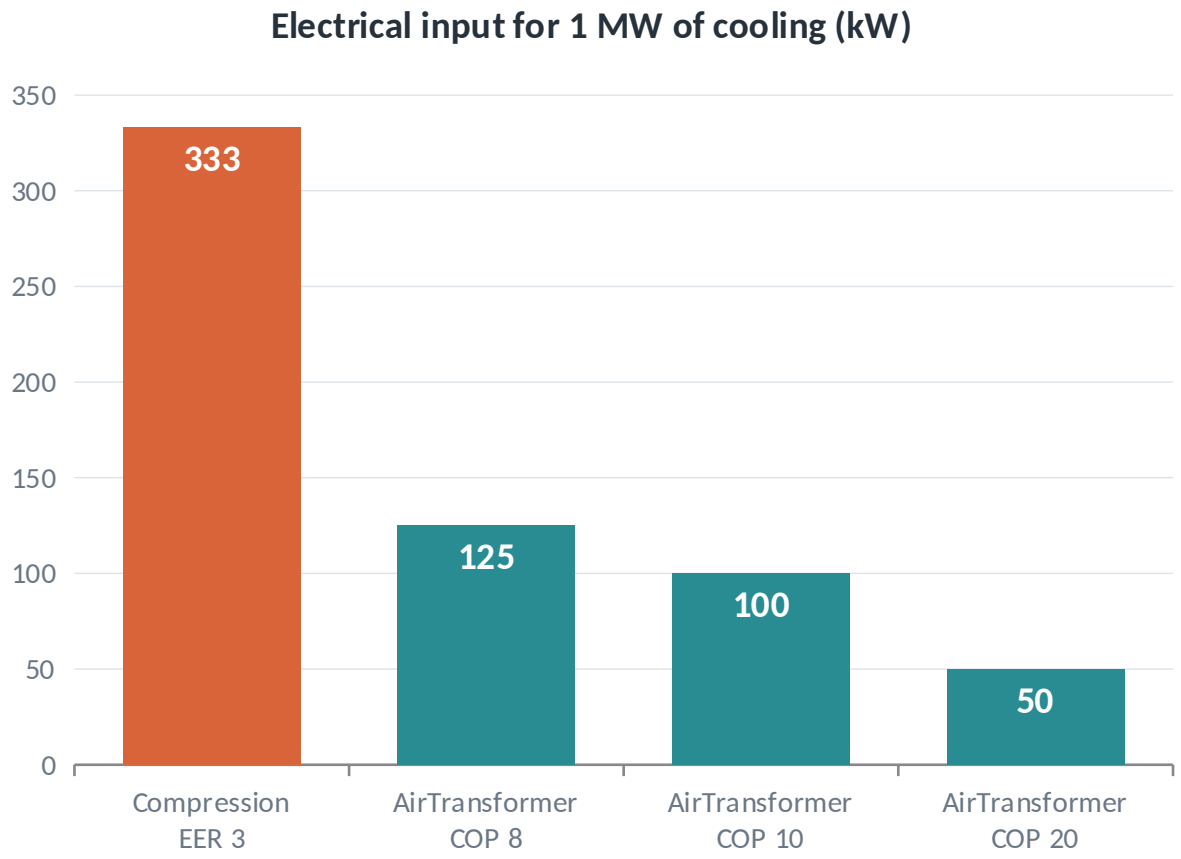
Compared at equal useful cooling, including all auxiliaries.

Alternative	Limit for the operator	AirTransformer
Compression cooling with free cooling	Free-cooling hours fall in a heatwave; compressor input peaks at EER ≈ 3 (333 kW per MW)	Waste heat replaces compressor work: 50–125 kW per MW at installed COP 8–20
Desiccant systems with mechanical post-cooling	Drying still needs a vapour-compression stage to deliver the cooling	Staged adsorption cooling delivers the duty without vapour compression
Closed-cycle adsorption chillers (silica gel, zeolite)	Vacuum vessels make chilled water that still needs air handling	Open-cycle contactors treat the air directly at atmospheric pressure
Atmospheric water harvesting or dehumidification (e.g. AirJoule)	Built to produce water or dry air, not to carry a facility cooling duty	Built around the cooling duty; water recovery is an operating mode

§ The staged MOF exchanger with integrated heat removal is the subject of Savery's 2025 family FR2508386 and its PCT extension.

Per MW: 208–283 kW of grid headroom in a heatwave

Engineering comparison against compression at EER 3, at 8,000 effective hours per year.



208–283 kW

grid capacity released per MW of cooling

1.7–2.3 GWh

electricity saved per MW each year

4,800–7,200 m³

external water saved per MW each year

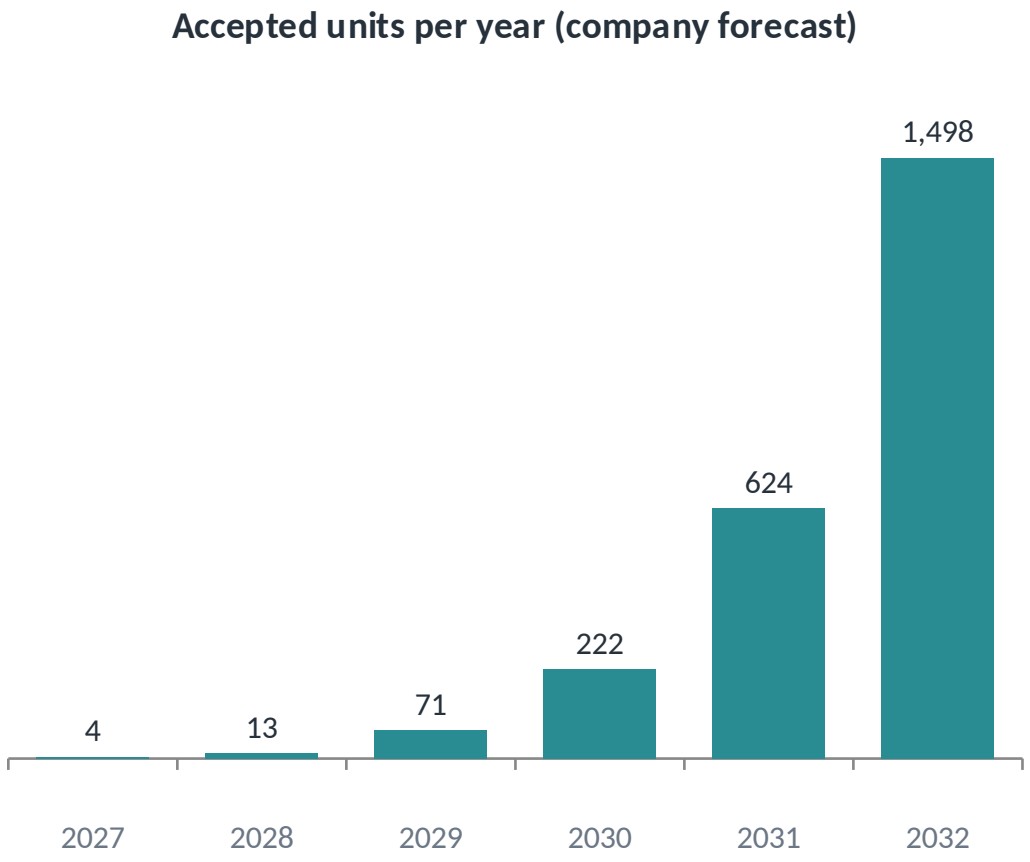
25–35 months

customer payback at €0.25/kWh; 35–48% cash ROI a year

C05 outcome: urban cooling that keeps its duty in extreme heat with less electricity and water, verified by matched-service tests at M24.

Sold on acceptance, grown by replication

€120k per unit: €110k equipment, €8k installation, €2k engineering; then maintenance at 3% a year.



Revenue model

Billing 20% on order, 20% on completion, 55% on acceptance, 5% after six months. Maintenance from month 12: 60% attachment, 90% renewal.

Acquisition and replication

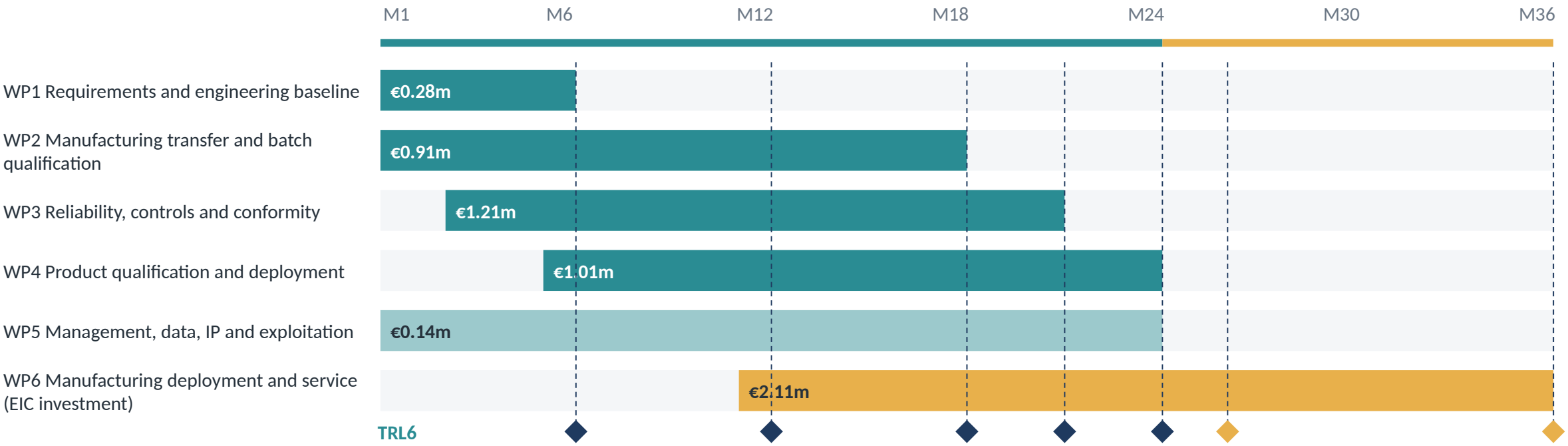
Pods of one account executive and one application engineer: eight qualified accounts a year, half convert after a 12-month validation. Each acceptance leads to more units on site, then further sites and regions.

Channel ecosystem

Design offices, installers, distributors, maintainers, integrators and site operators (e.g. Cap Ingelec, Rexel, Dalkia, HLRS, Cineca). Discovery so far: 60 conversations, 20 in-depth.

TRL6 machine to qualified TRL8 product in 24 months

April 2027 – March 2029 · 258 person-months · €3.55m eligible cost · seven evidence gates



M6
Specification, verification protocol and test plan approved

M12
Drawings and process window released; rights disposition recorded

M18
Six batches, 12 contactors qualified; 1,200 independent fixture-hours

M21
4,000 endurance hours; conformity file and software released

M24 · TRL8
1,600 field hours; ≥60% less electricity, ≥40% less water verified

M26
Serial tooling and manufacturing partner qualified

M36
Five installation and service packages delivered

The team to build, qualify and sell the product

Core team full-time, two engineers in post, eleven roles planned against work-package needs.



Frédéric Cui

Founder and CEO, technology and product

Sorption and thermal modelling, prototype validation, industrial transfer; TU Eindhoven, ARMINES



Qin Yu

Engineering and production lead

HVAC installation, supplier qualification, commissioning, QHSE



Ina Wang

Commercial lead

15+ years B2B: pricing, enterprise accounts, cross-border sales



Nicolas Paris

Interim CFO (30%)

Financial planning, energy markets, investment funds

Company board



Christian Houzé
Engineer of the Corps des Mines; international scientific and industrial experience



François Giger
Engineer of the Corps des Mines; international scientific and industrial experience



Olivier Labbé
Data-centre design, construction and commercialisation

Scientific board

NS

Prof. Norbert Stock
Kiel University, Institute of Inorganic Chemistry: CAU materials

AA

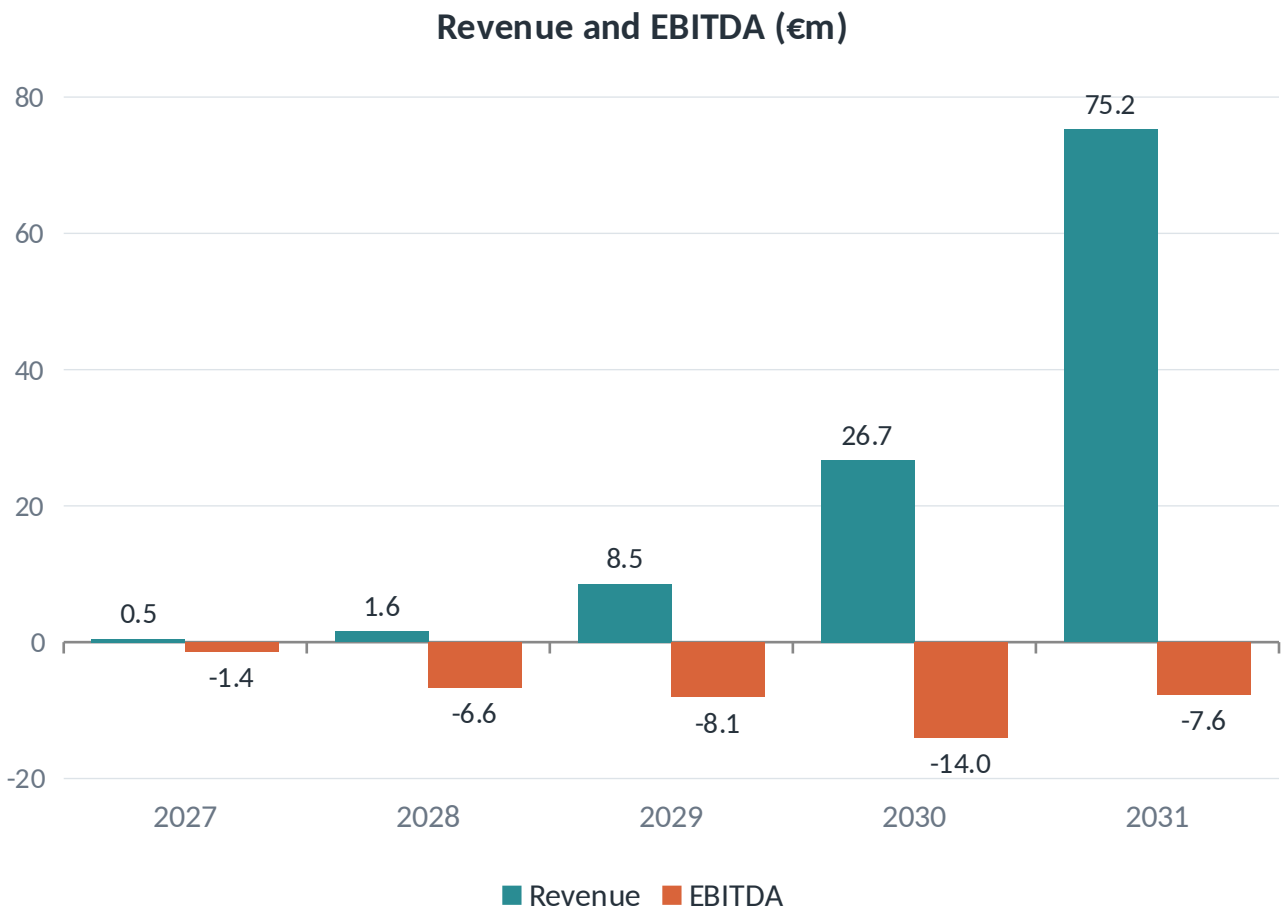
Dr Aline Auroux
Emeritus CNRS Research Director, IRCELYON: adsorption calorimetry



Driss Stitou
Solid–gas sorption and thermal-energy conversion

€75m revenue in 2031 as factory capacity scales

Company forecast, nominal €m. The EIC grant co-finances costs already in the business plan.



Funding path

Round	Timing	Amount
Seed, incl. €2m EIC	Sep–Nov 2027	**€8m**
Series A: pilot factory	April 2028	**€25m**
Series B: replication	January 2030	**€70m**
Series C: regional capacity	January 2032	**€35m**

€95m cash needed to reach operating break-even; Seed, A and B raise €101m net by 2030

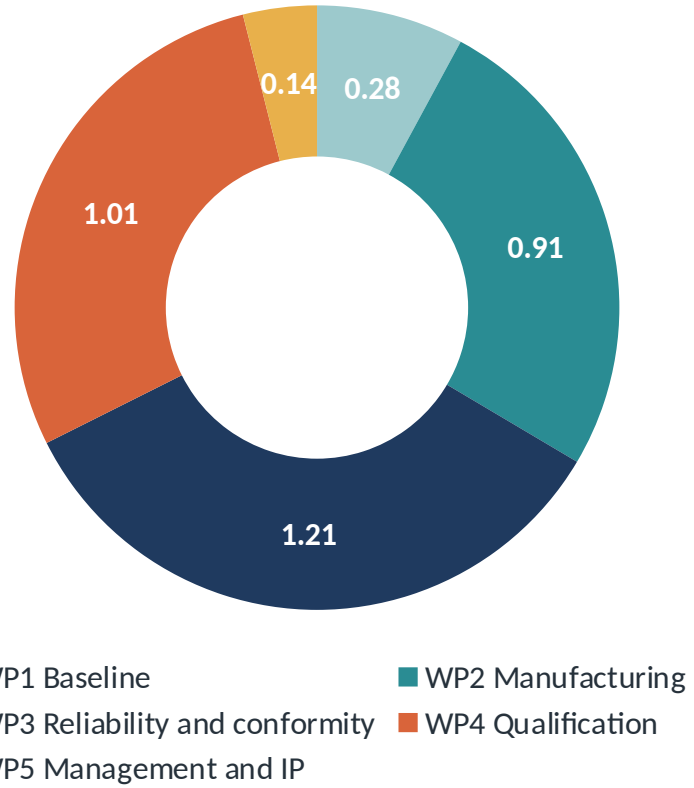
€181m 2032 revenue, 1,498 accepted units

> 0 cash balance in every month of the plan

The ask: €4.49m to qualify the product and scale

€2.49m grant (70% of €3.55m) + €2m EIC investment anchoring an €8m Seed, closing November 2027.

Grant action: €3.55m eligible cost by work package (€m)



€2.49m

EIC grant

70% of the €3.55m productisation action: manufacture, reliability, conformity and field qualification to TRL8

€2.0m

EIC investment

€1.9m to manufacturing deployment and service (tooling, partner, application engineering, first packages); €0.1m to action co-financing

€6.0m

Other Seed investors

Launch September 2027, close November 2027; co-finances the action and the company's wider growth plan

Why the EIC: qualification risk sits before revenue. Blended finance retires it, and EIC equity signals that to the industrial investors of the Seed and Series A.

Waste heat in, resilient cooling out

Why now

Heatwaves, grid limits and PUE rules are hitting data centres now, and liquid cooling has made 50–70 °C rack heat available at scale.

Why us

Ten years from proof of concept to a complete TRL6 machine at COP 20; Savery-owned IP; a team covering science, engineering and sales.

Why the EIC

€4.49m of blended finance turns the machine into a qualified TRL8 European product and anchors the €8m Seed.



Frédéric Cui · founder and CEO, Savery Tech · AirTransformer

Backup: rating basis and customer economics

For questions on the COP figures and the payback range.

Rating basis

- Machine COP 20: delivered sensible cooling ÷ electrical input, 60 °C waste heat, 2025 EN test.
- The EN heat loops ran at static pressure, so the circulating pump was outside the test input.
- Installed COP 8–20: site pumps, fans, controls and water treatment included.
- Regeneration heat and electrical COP are reported separately at every rating point.

ΔP = Q × (1/EER ref − 1/COP installed)

Scenario per 100 kW of cooling	Net saving / yr	Payback	Cash ROI / yr
€120k installed, €0.25/kWh (main)	€42–58k	25–35 months	35–48%
€120k installed, €0.20/kWh	€33–46k	31–43 months	28–39%
€100k installed, €0.20/kWh (future cost)	€33–46k	26–36 months	33–46%

Assumptions: 100 kW useful cooling; 8,000 effective hours a year of displaced compression; reference EER 3; installed COP 8–20; waste heat at zero marginal cost; baseline external water 1.5 L/kWh of cooling at €4/m^3 with a 40–60% make-up reduction (€1,920–2,880 a year); incremental service €2,000 a year.

Electrical headroom and cooling continuity add value beyond the cash saving.

Backup: delivery risks and responses

Each risk has an owner and a review point in the Implementation Plan.

Risk	Response and owner	Review point
Recruitment delays on the 213 planned-role PM	Frédéric and Qin prioritise M1 posts, assign deputies, rephase dependencies; scoped specialist services	Before M1, then monthly
CAU processing reduces output, consistency or life	Batch traceability, after-shaping tests, ageing and contamination checks, quarantined lots	Batch releases; M12 and M18
A rating point does not transfer to a site	Verify source, sink and auxiliaries; revise the released operating domain; repeat the matched-service test	M6 and each campaign
Conformity or third-party rights need a design change	Quality lead and Frédéric obtain the assessment, record the design change or rights, close findings	Before each release
Financing or grant payments arrive late	Phase hiring, procurement and deployment against available funds	Monthly cash review
Orders or acceptances arrive later than planned	Ina leads account qualification, reference-site evidence and staged offers	Quarterly pipeline review